

Exercises for Section 10.3

Directions: For exercises 1 - 12, determine the vertex, focus, and directrix of the parabola.

1. $x = 8y^2$

2. $y = \frac{1}{4}x^2$

3. $y = -4x^2$

4. $x = \frac{1}{8}y^2$

5. $x = 36y^2$

6. $x = \frac{1}{36}y^2$

7. $(x - 1)^2 = 4(y - 1)$

8. $(y - 2)^2 = \frac{4}{5}(x + 4)$

9. $(y - 4)^2 = 2(x + 3)$

10. $(x + 1)^2 = 2(y + 4)$

11. $(x + 4)^2 = 24(y + 1)$

12. $(y + 4)^2 = 16(x + 4)$

Directions: For exercises 13 - 19, graph the parabola, labeling the focus and the directrix.

13. $x = \frac{1}{8}y^2$

14. $y = 36x^2$

15. $y = \frac{1}{36}x^2$

16. $y = -9x^2$

17. $(y - 2)^2 = -\frac{4}{3}(x + 2)$

18. $-5(x + 5)^2 = 4(y + 5)$

19. $-6(y + 5)^2 = 4(x - 4)$

Directions: For exercises 20 - 25, find the equation of the parabola given information about its graph.

20. Vertex is $(0,0)$; directrix is $y = 4$, focus is $(0, -4)$.

21. Vertex is $(0,0)$; directrix is $x = 4$, focus is $(-4,0)$.

22. Vertex is $(2,2)$; directrix is $x = 2 - \sqrt{2}$, focus is $(2 + \sqrt{2}, 2)$.

23. Vertex is $(-2,3)$; directrix is $x = -\frac{7}{2}$, focus is $(-\frac{1}{2}, 3)$.

24. Vertex is $(\sqrt{2}, -\sqrt{3})$; directrix is $x = 2\sqrt{2}$, focus is $(0, -\sqrt{3})$.

25. Vertex is $(1,2)$; directrix is $y = \frac{11}{3}$, focus is $(1, \frac{1}{3})$.